ECE 3441 Spring 2022
Quiz 1

VERSION B

- Closed book, closed notes
- No cheat sheet is allowed.
- Only four function calculators are allowed.
- Show all details of your work to obtain credit (The answer alone will not suffice).
- You may not talk during the quiz to others or to yourself.
- You have 50 minutes.

Please do not open booklet unless instructed to do so, or substantial penalty will result.

First Name Last Name, Peoplesoft ID, Date *DD/MM/YYYY*

Question	Points	Maximum
1		20
2		25
3		25
4		20
5		20

Total: _____/100 possible points + 10 bonus points

Question #1:

a) Convert the following decimal value to its corresponding binary value:

$$366.75_{10}$$

b) Do the following 4-bit binary arithmetic and indicate whether there is overflow (the binary numbers are not in 2s compliment):

$$\begin{array}{r} 0011 \\ \underline{-\ 1000} \end{array}$$

Question #2:

Simplify the following Boolean expression to a minimum sum of products and state the Boolean algebra laws used:

$$(AB' + C')(A + B')' + A$$

Question #3:

Prove that $(A \oplus B) \equiv (C \oplus D) = (A \equiv B)(C \equiv D) + (A \oplus B)(C \oplus D)$ using Boolean algebra.

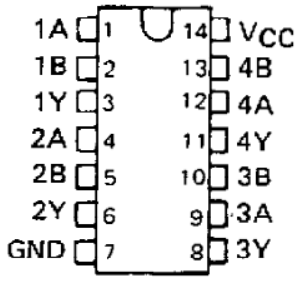
Question #4:

Find the minimum SOP of the incompletely specified function with the following truth table using only boolean algebra (no K-maps):

A	B	C	$F(A,B,C)$
0	0	0	1
0	0	1	0
0	1	0	X
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

Question #5:

Given the following datasheet and pinout for the SN74LS32N:



recommended operating conditions

		SN5432			SN7432			UNIT
		MIN	NOM	MAX	MIN	NOM	MAX	
V_{CC}	Supply voltage	4.5	5	5.5	4.75	5	5.25	V
V_{IH}	High-level input voltage	2			2			V
V_{IL}	Low-level input voltage			0.8			0.8	V
I_{OH}	High-level output current			−0.8			−0.8	mA
I_{OL}	Low-level output current			16			16	mA
T_A	Operating free-air temperature	−55		125	0		70	°C

electrical characteristics over recommended operating free-air temperature range (unless otherwise noted)

PARAMETER	TEST CONDITIONS†			SN5432			SN7432			UNIT
				MIN	TYP‡	MAX	MIN	TYP‡	MAX	
V_{IK}	V_{CC} = MIN,	I_I = −12 mA				−1.5			−1.5	V
V_{OH}	V_{CC} = MIN,	V_{IH} = 2 V,	I_{OH} = −0.8 mA	2.4	3.4		2.4	3.4		V
V_{OL}	V_{CC} = MIN,	V_{IL} = 0.8 V,	I_{OL} = 16 mA		0.2	0.4		0.2	0.4	V
I_I	V_{CC} = MAX,	V_I = 5.5 V				1			1	mA
I_{IH}	V_{CC} = MAX,	V_I = 2.4 V				40			40	μA
I_{IL}	V_{CC} = MAX,	V_I = 0.4 V				1.6			−1.6	mA
I_{OS}§	V_{CC} = MAX			−20		−55	−18		−55	mA
I_{CCH}	V_{CC} = MAX,	See Note 2			15	22		15	22	mA
I_{CCL}	V_{CC} = MAX,	V_I = 0 V			23	38		23	38	mA

a) What are the fan-in and fan-out of this chip?

_____ _____ _____ _____

b) What are the maximum values that can be used for pull-up and pull-down resistors?

_____ _____

_____ _____

Uniting theorems:

1. $XY + XY' = X$ 1D. $(X + Y)(X + Y') = X$

Absorption theorems:

2. $X + XY = X$ 2D. $X(X + Y) = X$

Elimination theorems:

3. $X + X'Y = X + Y$ 3D. $X(X' + Y) = XY$

Duality:

4. $(X + Y + Z + \cdots)^D = XYZ\ldots$ 4D. $(XYZ\ldots)^D = X + Y + Z + \cdots$

Theorems for multiplying out and factoring:

5. $(X + Y)(X' + Z) = XZ + X'Y$ 5D. $XY + X'Z = (X + Z)(X' + Y)$

Consensus theorems:

6. $XY + YZ + X'Z = XY + X'Z$ 6D. $(X + Y)(Y + Z)(X' + Z) = (X + Y)(X' + Z)$

Operations with 0 and 1:

1. $X + 0 = X$ 1D. $X \cdot 1 = X$
2. $X + 1 = 1$ 2D. $X \cdot 0 = 0$

Idempotent laws:

3. $X + X = X$ 3D. $X \cdot X = X$

Involution law:

4. $(X')' = X$

Laws of complementarity:

5. $X + X' = 1$ 5D. $X \cdot X' = 0$

Commutative laws:

6. $X + Y = Y + X$ 6D. $XY = YX$

Associative laws:

7. $(X + Y) + Z = X + (Y + Z) = X + Y + Z$ 7D. $(XY)Z = X(YZ) = XYZ$

Distributive laws:

8. $X(Y + Z) = XY + XZ$ 8D. $X + YZ = (X + Y)(X + Z)$

DeMorgan's laws:

9. $(X + Y)' = X'Y'$ 9D. $(XY)' = X' + Y'$

Exclusive OR

$X \oplus Y = X'Y + XY'$

Equivalence

$(X \equiv Y) = XY + X'Y'$

$$FO_L = \frac{I_{OL\,max}}{I_{IL\,max}}; \qquad FO_H = \frac{I_{OH\,max}}{I_{IH\,max}}$$

$$Fan\ out = \min\,(FO_L, FO_H)$$

$$R_{Pull\ up} < \frac{V_{CC} - V_{IH\,min}}{I_{IH\,max}}; \qquad R_{Pull\ down} < \frac{V_{IL\,max}}{I_{IL\,max}}$$